Somasekhar Thupakula

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PROFESSIONAL SUMMARY

Computer Science student with practical experience developing production-oriented software using Java, Spring Boot, PostgreSQL, Redis, and Docker. Built scalable REST APIs, implemented authentication and caching mechanisms, and designed concurrency-safe workflows through real-world projects. Seeking Software Engineer opportunities to contribute to building reliable, efficient, and maintainable software systems.

EDUCATION

Malla Reddy Deemed to be University(Hyderabad) 2023 — Present

Bachelor of Technology in Computer Science & Engineering (Expected 2027) CGPA: 7.9/10

SKILLS

- **Languages:** Java, Python, JavaScript
- **Backend:** Spring Boot, Spring Security, Spring AI, JPA/Hibernate, REST APIs, Microservices
- **DevOps & Tools:** Git, Docker, Linux, GitHub Actions, Maven, Vercel
- **Databases:** PostgreSQL, Redis, MinIO

PROJECTS

Synapse-Hub - Idea management for team GitHub

Tech Stack: Java 21, Spring Boot, PostgreSQL, Clerk OAuth, React, TypeScript, MockMvc, Docker

- Enforced multi-tenant isolation via PostgreSQL RLS and Spring Security filters, blocking cross-tenant IDOR access
- Built a dual-mode auth engine — Clerk OAuth2 and a mock JWT sandbox — cutting local setup to one Docker command
- Verified tenant isolation and API security via integration tests, hitting 100% pass across isolation, XSS, and rate-limiting suites

Distributed KV-Store GitHub

Tech Stack: Java 21, Netty, Junit, SLF4J, Docker, Docker Compose

- Networking: Architected a concurrent key-value store using Java 21 and Netty with a custom TCP wire protocol.
- Durability: Engineered crash-safe storage via write-ahead logging (WAL) disk flushes and atomic JSON snapshotting
- Replication: Implemented asynchronous primary-replica streaming with automatic client reconnection and lag tracking.

Natural Language Ticket Booking System GitHub

Tech Stack: Java, Spring AI, Supabase, SpringBoot, PostgreSQL, Redis, React, Docker

- Designed a two-tiered locking engine using Redis TTL soft holds and PostgreSQL pessimistic locks to eliminate double-bookings.
- Built a resilient transaction system with idempotent retries and automated compensating rollbacks for payment failures.
- Synchronized seat maps via Postgres CDC and built AI booking agent that routes directly through the transactional API.

ACHIEVEMENTS

- **Inspira** — Built a gamified food waste tracking application.
- **Vishesh** — Developed and deployed a full-stack employee management application using CRUD operations and Vercel deployment.